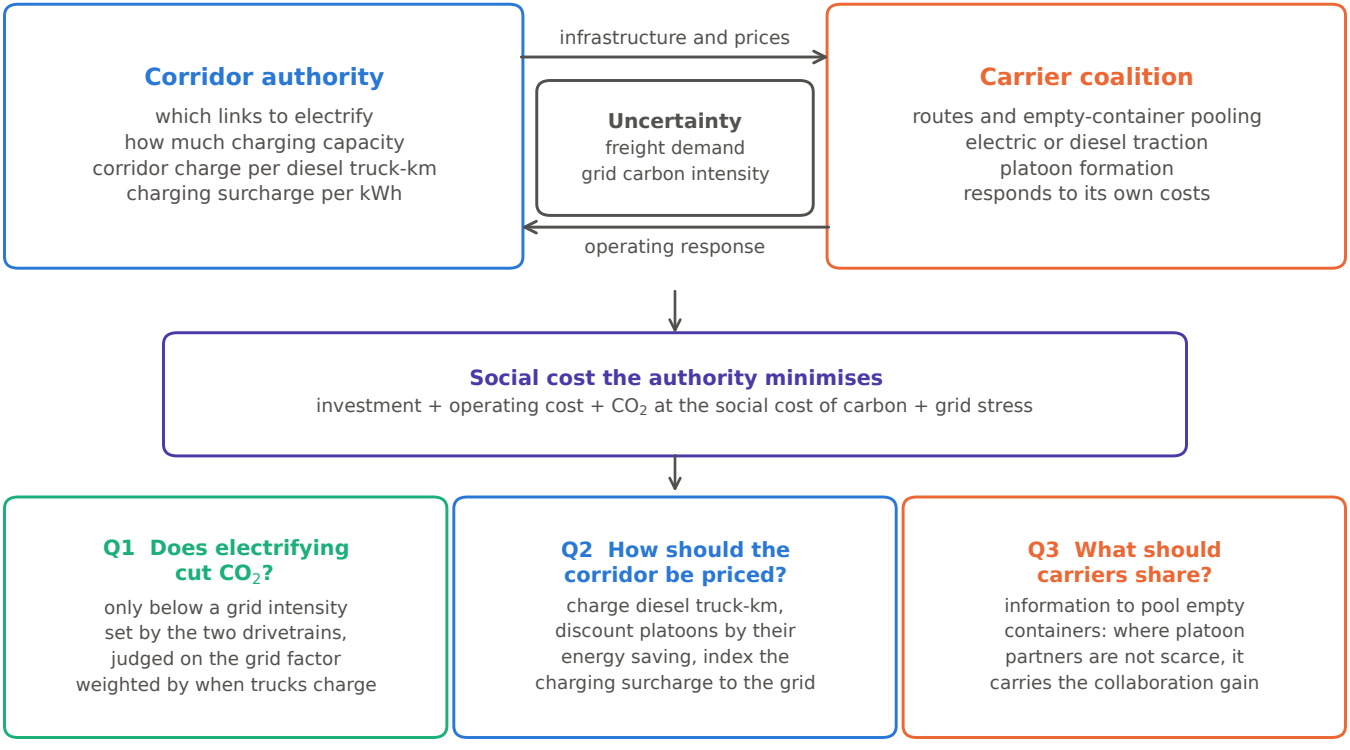
Corridor authority
which links to electrify
how much charging capacity
corridor charge per diesel truck-km
charging surcharge per kWh
infrastructure and prices
Uncertainty
freight demand
grid carbon intensity
operating response
Carrier coalition
routes and empty-container pooling
electric or diesel traction
platoon formation
responds to its own costs
Social cost the authority minimises
investment + operating cost + CO_2 at the social cost of carbon + grid stress
Q1 Does electrifying cut CO_2?
only below a grid intensity set by the two drivetrains, judged on the grid factor weighted by when trucks charge
Q2 How should the corridor be priced?
charge diesel truck-km, discount platoons by their energy saving, index the charging surcharge to the grid
Q3 What should carriers share?
information to pool empty containers: where platoon partners are not scarce, it carries the collaboration gain